

FUTURE SCOPE DETAILED

1.1 Offline-First Marine Navigation

A potential future extension of ORCA is an **offline-first marine navigation capability** designed for operations in areas where mobile network connectivity becomes intermittent or unavailable. The proposed approach would allow the application to continue providing navigation-related functionality through the device's GNSS receiver, locally stored map data and locally stored route information. The proposed system may use **GNSS as the primary positioning source**, while maintaining locally stored route points and navigation information for operation without continuous internet connectivity. Connectivity-dependent services such as weather updates, AIS information and backend synchronisation may resume when network connectivity becomes available.

1.2 GNSS-Based Offline Positioning

The proposed navigation extension may utilise the phone's built-in **GNSS receiver** for determining:

- Latitude and longitude
- Speed
- Bearing
- Altitude
- Position accuracy
- Distance travelled

GNSS-based positioning can operate independently of mobile data or SIM connectivity. This would allow location tracking and distance calculation to continue even when cellular connectivity is unavailable.

1.3 GNSS Quality Filtering and Position Reliability

To improve the reliability of offline navigation, a potential future location engine may incorporate a filtering layer.

The proposed filtering mechanism may:

- Reject low-accuracy position readings.
- Detect and reject impossible position jumps.

- Perform speed sanity checks.
- Accumulate travelled distance only from accepted position points.
- Store position, timestamp, accuracy, speed, bearing and source information locally.

A confidence indicator may also be explored to communicate the reliability of the current position to the user.

Proposed status indicators

- **GNSS Locked** — High confidence
- **Offline / Estimating** — Medium confidence
- **Low Confidence**
- **No Reliable Position**

1.4 Short-Term Sensor-Assisted Navigation

A further potential extension is the use of the device's **accelerometer, gyroscope and magnetometer** for short-term navigation assistance.

The proposed sensor-assisted approach may be considered for:

- Maintaining navigation information during temporary GNSS signal degradation.
- Detecting changes in vessel direction.
- Supporting heading estimation.
- Smoothing the displayed movement track.

The proposed use of inertial sensors would be limited to short-duration assistance, after which GNSS could be used again to correct the estimated position.

1.5 Offline Marine Maps

ORCA may be extended to support **offline marine map tiles** through locally stored map data.

A potential implementation may use:

- Flutter map
- MBTiles or an equivalent local tile-storage mechanism

The objective would be to allow the map interface and locally available navigation information to remain accessible without continuous internet connectivity.

1.6 Local Route Storage and Synchronisation

A future version may provide local storage of marine routes and navigation records.

The proposed local storage may contain:

- Route points
- Latitude and longitude
- Timestamp
- Accuracy
- Speed
- Bearing
- Navigation source

These records could remain available offline and may be synchronised with the backend when connectivity becomes available again.

A local database such as **SQLite or an equivalent local storage solution** may be explored for this purpose.

1.7 Offline Route Guidance

The future navigation module may provide route guidance based on previously defined or user-selected waypoints.

Potential navigation information may include:

- Next waypoint
- Distance to waypoint
- Recommended heading
- Estimated arrival time
- Cross-track information
- Direction guidance based on heading difference

This functionality is proposed as an extension of ORCA's existing map and location capabilities.

1.8 External GPS / Marine Equipment Integration

Another proposed extension is integration between the ORCA mobile application and **external marine navigation equipment** available on boats.

Potential external sources include:

- Boat-mounted GPS
- GPS-enabled VHF equipment
- AIS equipment
- External GPS modules

The proposed communication methods may include:

- Bluetooth
- Wi-Fi
- Wired/USB connection where supported

Marine equipment providing position information through **NMEA 0183** may be considered as an external data source for the navigation system.

The objective would be to allow ORCA to use an external positioning source where available rather than relying exclusively on the phone's internal GNSS receiver.

1.9 Connectivity-Aware Marine Operation

Because cellular connectivity may become unreliable farther offshore, ORCA may be extended with a **connectivity-aware architecture**.

The proposed model may operate through multiple levels:

When internet is available:

Live weather, backend services, AIS-related information and synchronisation.

When internet is unavailable:

GNSS-based positioning, offline map, locally stored route and locally available navigation information.

When connectivity returns:

Synchronisation of locally stored navigation and route information.

1.10 Boat-to-Boat Collaborative Communication

A potential future innovation is a **local boat-to-boat communication layer**.

When boats are within communication range, the system may explore local exchange of:

- Last-known positions
- Route information
- Safety information
- Relevant alerts

Possible communication mechanisms under consideration include **Bluetooth, Wi-Fi-based local communication and VHF-supported communication**, depending on the available equipment and deployment conditions.

This would be explored as a complementary communication mechanism for situations where direct internet connectivity is unavailable.

1.11 Integration with Indian Government Marine Systems

ORCA may potentially be designed to **co-exist or integrate with existing Indian marine communication and tracking systems**, particularly government-supported systems such as **VCSS and Nabhmitra**, where technical access and integration mechanisms are available.

The proposed integration could potentially enable ORCA to make use of:

- Vessel position information
- Two-way communication
- SOS / emergency communication
- Weather and cyclone warnings
- PFZ-related information
- IMBL geofencing
- Navigational assistance

This is proposed as an interoperability and integration direction rather than a claim of current implementation.

1.12 Satellite-Based Communication for Offshore Operations

For operations beyond normal cellular connectivity, a future ORCA architecture may explore integration with **satellite-based communication systems**.

The proposed objective would be to maintain communication and emergency information exchange when terrestrial mobile networks are unavailable.

This may be particularly relevant for fishing vessels operating farther from the coast and for emergency scenarios where conventional internet connectivity cannot be relied upon.

1.13 Emergency and Safety Extensions

Future versions of ORCA may explore additional emergency-oriented functions such as:

- SOS with last-known position
- Emergency message storage
- Emergency contact information available offline
- Local safety alerts
- Position sharing when communication becomes available

The purpose would be to preserve critical safety information even during temporary loss of connectivity.

1.14 Fuel and Route Analysis

Based on the requirements identified during the fisherman interaction, ORCA may further explore route planning that considers **distance, fuel usage and vessel-related parameters**.

Potential outputs may include:

- Route distance
- Estimated fuel requirement
- Waypoint distance
- Return-route information
- Route comparison

These capabilities are proposed future extensions and are not presented as completed implementations.

1.15 Future Scope for Marine Vessel Awareness

A further potential extension is integration of vessel-position information to improve awareness of nearby vessels.

This could involve future integration with **AIS or compatible external vessel-tracking sources**, subject to data availability and integration access.

The objective would be to provide an additional layer of vessel awareness alongside the existing map and navigation functions.

1.16 Consolidated Future Innovation

The proposed future evolution of ORCA can therefore be represented as:

Current ORCA



Offline GNSS Navigation



Sensor-Assisted Positioning



Offline Marine Maps



Local Route Storage



External GPS / VHF / AIS Integration



Boat-to-Boat Communication



Satellite / Government-System Integration



Extended Emergency & Marine Safety Intelligence